Hypothesis Testing Basics

"Suppose a company claims that its average delivery time is 30 mins. How can you check whther this claim is actually true using data?"

Hypthesis testing is a statistical method used to check a claim about a population using sample data.

Technically:
It is a statistical inference procedure that evaluates evidence from sample data to decide whether a null hypthesis should be rejected.

Null Hypthesis (H0)
The null hypthesis is the starting assumption that there is no change, no difference, no effect.

Techincally:
H0 is a statistical statement about a population parameter that is evaluated using sample evidence

Example: A company claims that its average delivery time is 30 minutes,
H0:μ=30

Other Examples:
1. Manufacturing: Defect rate remains 5%.
2. Marketing: A new campaign has not changed conversion rate.
3. Healthcare: A new treatmet has no imporovement over the existing one.

Interview Point:
We do not prove H0 is true. We can either reject it or fail to reject it.

Alternative Hypthesis(H1)
The alternative hypthesis represents the effect, difference, or change we want to find evidence for.

Technically:
H1 specifies outcomes inconsistent with the null hypthesis.

For delivery Example:
H0:μ=30
H1:μ<>30
This tests whether the average delivery time differs from 30 minutes in either direction.

Business Questions

Has delivery time changed?	H1:μ<>30
Has average sales increased?	H1:μ>1000
Has the defect rate decreased?	H1:p<0.05

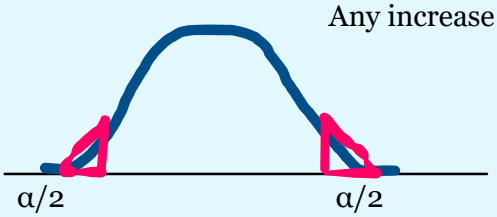
The research question determines the alternative hypthesis. Don't choose the direction after seeing the result.

One tailed vs Two tailed Tests

A **one tailed test** examines a difference in one specified direction. A **two tailed** test examines difference in either of the direction.

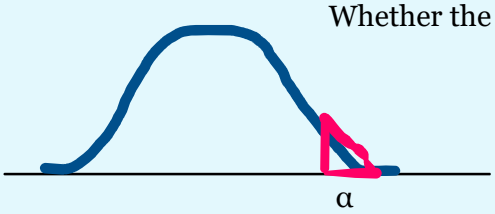
Two tailed test
H1:μ!=30

Any increase or decrease



Right tailed test
H1:μ > 30

Whether the mean increased.



Business Examples:
Two tailed : Has a website redesign changed conversion rate?
Right tailed: has a new campaign increased sales?
Left tailed: Has a process improvement reduced the delivery time.

Level of Significance(α)

The significane level is the maximum probability of a Type 1 error we are willing to accept under testing procedure.

Significance Level		Confidence Level Equivalent		
α=0.10		90%		
α=0.05		95%		
α=0.01		99%		

At α=0.05, we accpet a 5% long Type 1 Error rate if the null hypthesis is true and the test assumptions hold.

Business Example: A Pharmaceutical company may choose stricter significance level than a team conducting an initial medical experiment because the cost of false positive is different.

A 5% significance level does not mean there is 5% probability that the null hypothesis is true

Test Statistic

A test statistic measure how far the observed sample result is from what we would expect of the null hypothesis were true.

For one sample Z-test for a mean when population Standard deviation is known:

$$Z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Where:
Xbar = observed sample mean
μ0 = mean assumed under H0
σ = population standard deviation
n = Sample Samize

Example:
A delivery company claims the mean delivery time is 30 minutes.
Hypothesized mean = 30 minutes
Sample Mean = 32 minutes
Population SD = 6 minutesd
Sample Size = 36
Significance level = 0.05

$$SE = \sigma/SQRT(n) = 6/SQRT(36) \qquad 1$$

$$Z = (32-30)/1$$
$$2$$

For a two tailed test at α=0.05. the critical values are approx -1.96 and +1.96
Since Z = 2 is beyond +1.96, reject H0.

Interpretation
The sample provides statistically significant evidence that the population mean differs from 30 minutes.

P- Value

The p-value tells us how unusual the observed result, or more extreme result, would be if the null hypothesis were true.

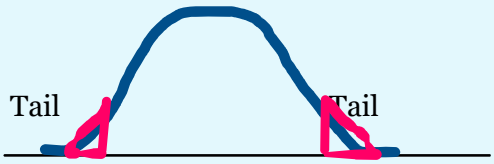
Techincally:
It is the probability, assuming H0 and statistical model, of obtaining a test statistic at least as extreme as the observed statistic in the firection or directions specified by H1.

For the two tailed Z- test above:

$$p\text{-value} \approx 0.0455$$

0.0455 0.05
Decision: 0.0455< 0.05
Reject H0.

Standard Normal Curve



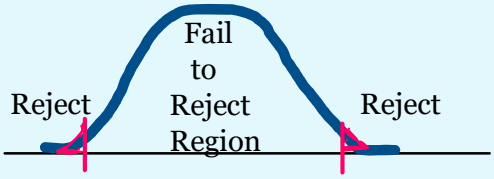
Remember:
Small p-value: evidence against H0
Large p-value: insufficienct evidence to reject H0.
The p value is not the probability that H0 is true.

Critical value Approach

A critical value is a threshold that seperates the rejection region from the non- rejection region.

For a two tailed Z- test at α=0.05:

$$Z_{critical} = \pm 1.96$$



For Z = 2, which lies beyond +1.96, Therefore rejects H0

Decision rule

Two equivalent approaces for the same test

P - Value Approach	Critical Value Approach
If p≤α, reject H0.	If the statistic lies in the rejection region, reject H0
If p>α, fail to reject H0.	Otherwise fail to reject H0.

Use the delivery time example:

Language: "Fail to reject H0" does not mean " accept or prove H0". It means the available evidence is isufficient to reject it.

Type I and Type II Errors

Type I Error (α) False Positive

Rejecting a true null hypthesis.

Example: Concluding that a new campaign improves sales when it actually does not.

Type II Error (β) False Negative

Failing to reject a false null Hypothesis.

Example: Missing a genuinely effective campaign because the sample provides insufficient evidence.

Power Of a Test

Statistical power is the probability of correctly rejecting a false null hypothesis.

$$Power = 1-\beta$$

Power generally increase when:

1. Sample Size increases
2. The true effect is larger
3. Measurement variability decreases.
4. The significance level increases, all else equal.

Business Example:

A company testing two website designs needs enough visitors to detect meaningful difference in conversion rate.

A statistically significant result is not automatically a a large or valuable busines improvement.

Statistical vs Practical Significance

Statistical Significance.
Is the observed result sufficiently inconsistent with the null hypothesis?
Practical Significance:
Is the size of the effect important enough to matter in the real world?

Example: A website redesign increases conversion rate from 5% to 5.05%. With a very large sample, this tiny difference might be statistically significant, yet the added revenue not justify the development cost.

You have to consider the effect size, implementation cost, business benefit, risk- not just and the p value